

costs money. Imagine Ethereum as one giant computer with many applications (i.e., smart contracts). If people want to use the computer, they must pay for each unit of computation. A simple computation such as sending ether (ETH) requires minimal work to update a few account balances and thus has a relatively small gas fee. A complex computation involving minting tokens and checking various conditions across many contracts requires more gas and thus will have a higher fee. The gas fee may lead to a poor user experience, however. It forces agents to maintain an ETH balance to pay it and leads to worry about overpaying, underpaying, or the transaction not taking place at all. So initiatives are ongoing to eliminate gas fees from end users. There are also competitor chains that completely remove this concept of gas.

However, gas is a primary mechanism for preventing system attacks that generate an *infinite loop* of code. It is not feasible to identify malicious code of this kind before running it, a problem formally known in computer science as *the halting problem*. Suppose a car is on autopilot, stuck in full throttle with no driver. Gas acts as a limiting factor: the car will stop eventually when the gas tank empties. In the same way, gas fees secure the Ethereum blockchain by making such attacks cost-prohibitive. They incentivize highly efficient smart contract code since contracts that use fewer resources and reduce the probability of user failures have a much higher chance of being used and succeeding in the market.